Staff answers and worked costs

Design and Technology LAUNCH Spring 1 Week 1 Thursday 7 January 2027 P1 Year A

Use these models to check reasoning. The final plan and justification belong to the pupil.

Model arithmetic

Ingredient	Amount	Cost
Wholemeal wrap	1 wrap	£0.35
Ready-to-eat cooked beans	80 g drained	£0.30
Carrot	60 g	£0.15
Cabbage	40 g	£0.10
Plain yoghurt	60 g	£0.20
Apple	1 small	£0.30
Total used		£1.40
Budget left		£0.60

35p + 30p + 15p + 10p + 20p + 30p = 140p = £1.40. 200p − 140p = 60p.

Worked balance reason

My £1.40 meal has starchy food from the wrap, protein from beans, vegetables and fruit, and dairy from yoghurt. It leaves 60p of my £2.00 budget.

Beans count in the protein group in this example. Dairy may be served alongside the wrap. Oils do not need to be added solely to complete a group.

A checked substitution

Apple to pear: 140p − 30p + 35p = 145p (£1.45). Money left: 55p. The fruit group is retained.

Yoghurt to the listed soya alternative: 140p − 20p + 30p = 150p (£1.50). Money left: 50p. Check calcium fortification, dietary suitability and actual labels first. This is not automatically an allergy-safe swap.

Staff checks and misconceptions

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Hinge and response

Correct choice: 60p. £1.40 is the amount spent; £3.40 adds the budget to the cost. Rehearse 200p minus 140p if needed. Parallel check: 200p minus 155p leaves 45p.

A balanced dish needs equal food group sections.

Balance is built across the day or week. Groups have different proportions and oils are used in small amounts.

A portion price is the price of a full pack.

This card charges only for the stated amount used. Buying whole packs may cost more.

Changing an ingredient means adding a second full cost.

Remove the old cost before adding the new one.

Any beans can go straight into a cold wrap.

Only cooked beans labelled ready to eat are used here. Dry beans require a separate cooking process.

Cold call with choice of response

Foundation: Point to the protein food. Expected: ready-to-eat beans.

Middle: Why do we subtract the apple first? Expected: its cost is removed when it is replaced.

Higher: What can invented portion prices hide? Expected: whole-pack purchasing costs, actual prices, waste and individual portion needs.

Evidence boundary

Accept a pupil-confirmed spoken, signed or scribed plan with accurate costs and a personal justification. Note help used. A copied model without pupil explanation does not establish the pupil’s reasoning. Do not record practical handling or tasting today.